

Lab 1

Convolution, DTFT, Z-Transform

Submission Format:

Each student must submit a single **ZIP file** named in the following format:

StudentName_Lab1_StudentID.zip

The ZIP file should contain:

1. A **PDF report** named: Report_StudentName_Lab1_StudentID.pdf
2. **MATLAB code files** corresponding to each problem, named in the format: ProblemX.m (where X represents the problem number).

Submission Contents:

- The PDF report should include MATLAB experiment results with corresponding plots/images.
- Each MATLAB code file should contain:
 - The complete solution for the corresponding problem.
 - Properly structured and commented in English for clarity.

Submission Email:

Send your submission to: dsp_speit2026@163.com

Deadline:

April 15th, 12:00 PM (Noon)

You are developing a smart audio processing system for music production, speech enhancement, and signal analysis. Your task is to verify the system's performance through convolution calculations (for reverberation effects), spectral analysis (DTFT), and filter stability analysis (Z-transform).

Part 1: Convolution - Simulating Audio Reverberation

In music production, reverberation is essential for making sounds more natural. It can be modeled using the impulse response (IR) of a room, which characterizes how sound reflects and decays over time.

Problem 1: Simulating a Recording Studio's Reverberation

The impulse response $h[n]$ of a small recording studio is given by:

$$h[n] = \delta[n] + 0.5\delta[n-1] + 0.3\delta[n-2]$$

where $\delta[n]$ is the unit impulse function.

A simple drum sound signal is given as:

$$x[n] = [2, 4, 6, 4, 2, 0]$$

Compute the reverberated output signal:

$$y[n] = x[n] * h[n]$$

and plot both the **original** and **reverberated** signals.

Problem 2: Simulating Echo in an Open Space

Now, you need to simulate echo in an open outdoor environment, where sound waves are reflected from distant buildings.

The original sound source follows an exponentially decaying pattern:

$$x[n] = 0.8^n, \quad 0 \leq n \leq 10$$

i.e., $x[n] = 0.8^n(u[n] - u[n-11])$ with $u[n]$ being the unit step function.

Compute the **echoed signal**:

$$y[n] = x[n] * h[n]$$

and plot both the **original** and **echoed** signals.

Part 2: DTFT - Speech Signal Spectrum Analysis

In speech processing and communication systems, analyzing the frequency spectrum helps optimize speech coding, noise removal, and audio enhancement. For example, telephone speech is transmitted within a limited bandwidth, requiring frequency domain analysis.

Problem 3: Spectrum Analysis of a Speech Signal

A short speech segment can be modeled as:

$$x[n] = 0.8^n, \quad 0 \leq n \leq 10,$$

i.e., $x[n] = 0.8^n(u[n] - u[n-11])$.

Compute the Discrete-Time Fourier Transform (DTFT):

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

Plot both the **magnitude spectrum** $|X(\omega)|$ and **phase spectrum** $\angle X(\omega)$ for $\omega \in [-\pi, \pi]$.

Part 3: Z-Transform - Stability Analysis of a Filter

In audio processing and communication systems, filters are used to enhance or suppress certain frequencies. For example, high-pass filters help remove low-frequency noise, while low-pass filters smooth signals.

Problem 4: High-Pass Filter for Speech Enhancement

A high-pass filter used for **speech enhancement** is defined by the impulse response:

$$h[n] = (-0.5)^n u[n]$$

where $u[n]$ is the unit step function.

Compute its Z-transform:

$$H(z) = \sum_{n=0}^{\infty} (-0.5)^n z^{-n}$$

and determine its Region of Convergence (ROC).

Analyze:

1. Is this filter stable? (i.e., do all poles lie inside the unit circle?)
2. Would this filter be effective for speech enhancement? Why?

Problem 5: Stability Analysis of a Low-Pass Filter

A low-pass filter for **noise reduction** is given by:

$$h[n] = (0.7)^n u[n]$$

Compute its Z-transform and analyze:

1. The location of its poles
2. Whether the filter is stable
3. How does it compare to the high-pass filter from Problem 4?